

INSE 6710: Fundamentals and Applications of Cyber-Physical Systems

Assignment: Part 1

In this part of the assignment, we implement and evaluate a chi-squared anomaly detector for a networked control system based on a Continuous Stirred Tank Reactor (CSTR) model. The primary goal is to design a detector that achieves a 3% false alarm rate while ensuring stability and performance of the system. Simulations are run over 50 trials to validate the detection capability and analyze control behavior.

1. Chi-squared Detector Design Steps

To design the chi-squared detector, the following steps were followed:

- The residual is computed using the output measurement y and the estimated state $\hat{x}$.
- The test statistic $z(k)$ is computed as: $z(k) = r^t \Sigma^{-1} r$, where $r = y - C\hat{x}$ and Σ is the residual covariance.
- The threshold τ is chosen using the inverse chi-squared cumulative distribution function to achieve a 3% false alarm rate. With $\alpha = 0.03$ and residual dimension $v = 1$:
 $\tau = \text{chi2inv}(1 - 0.03, 1) = 4.7093$
- Detection occurs when $z(k) > \tau$.

2. Detector Design Parameters

Parameter	Value
Chi-squared threshold (τ)	4.7093
Residual covariance (Σ)	0.0016
False Alarm Rate Target (α)	3%
Residual dimension (v)	1

3. Simulation Results Over 50 Trials

The following table summarizes the performance metrics computed by averaging over 50 simulation runs:

False Positives (FP)	True Negatives (TN)	False Alarm Rate (%)	Avg. Tracking Error (Je)
750	19300	3.74%	2.6997

The detector closely achieves the desired false alarm rate of 3%, and the tracking error remains low, indicating stable control performance under no-attack conditions.

4. Resulting Plots

The following plots were generated to visually confirm the detector's behavior:

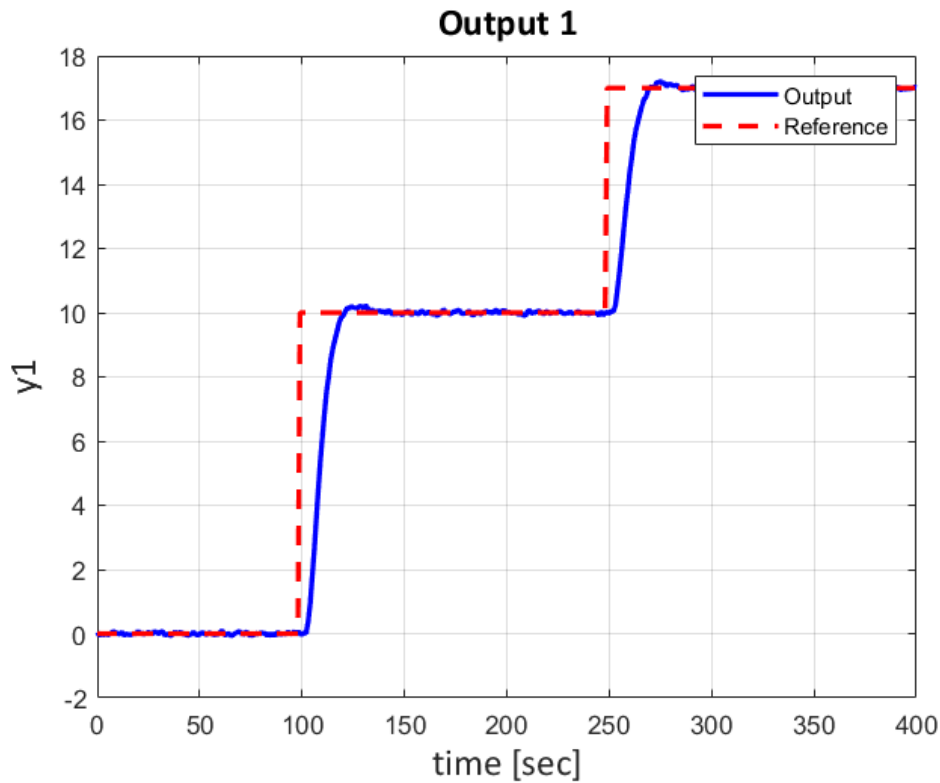


Figure 1: Output vs Reference

This plot compares the measured output $y(t)$ of the system with the desired reference signal $r(t)$. The output tracks the reference well at all three reference levels (0, 10, 17), showing proper system performance. Good tracking performance suggests that the LQI controller and Kalman estimator are correctly designed. Average tracking error $J_e = 2.6997$ is low, confirming accurate reference tracking across the time horizon. No instability or divergence, even during transitions.

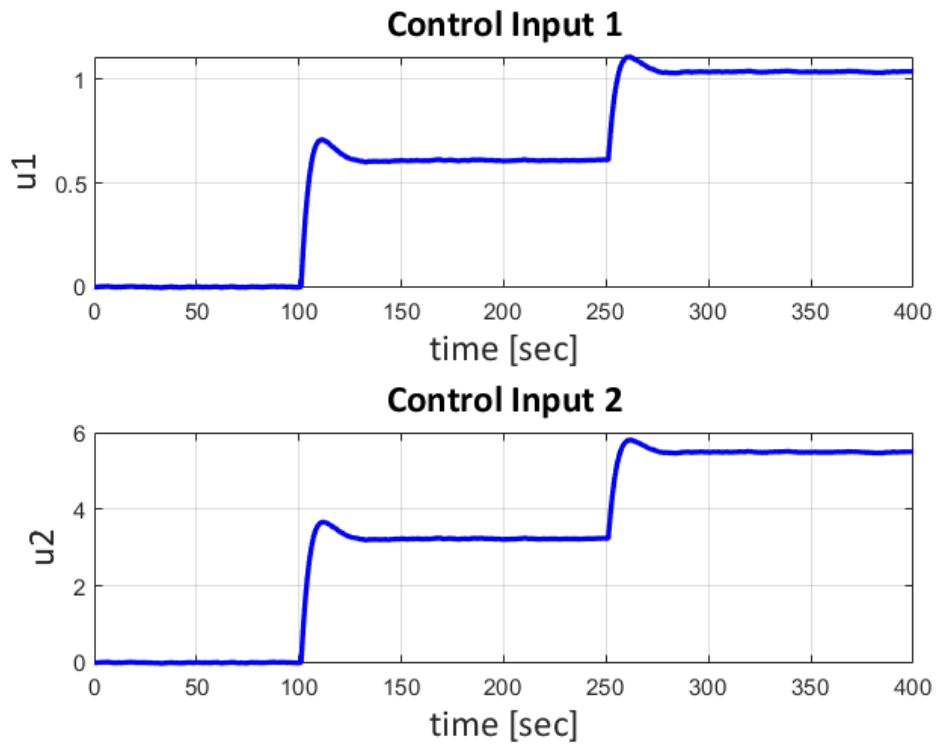


Figure 2: Control Inputs

This figure shows control inputs evolving over time in response to reference changes. Inputs are adjusting promptly and stabilizing smoothly after each setpoint shift. It is also indicating a well-tuned LQI gain. Moreover, it supports low tracking error J_e and stable output behavior.

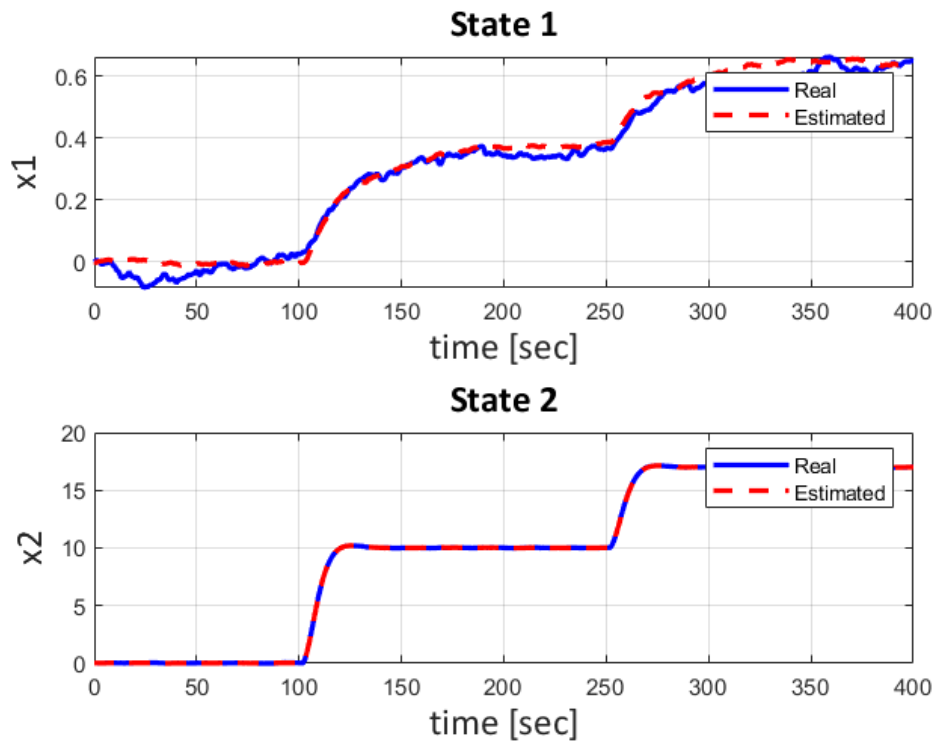


Figure 3: Real vs Estimated States

This figure compares the true plant states x_1 , x_2 with their estimates $\hat{x}_1$, $\hat{x}_2$ from the Kalman filter. State estimates are very close to real states throughout, indicating excellent observability and estimator performance. Ensures the chi-squared residuals $z(k)$ based on accurate predictions, improving detection reliability.

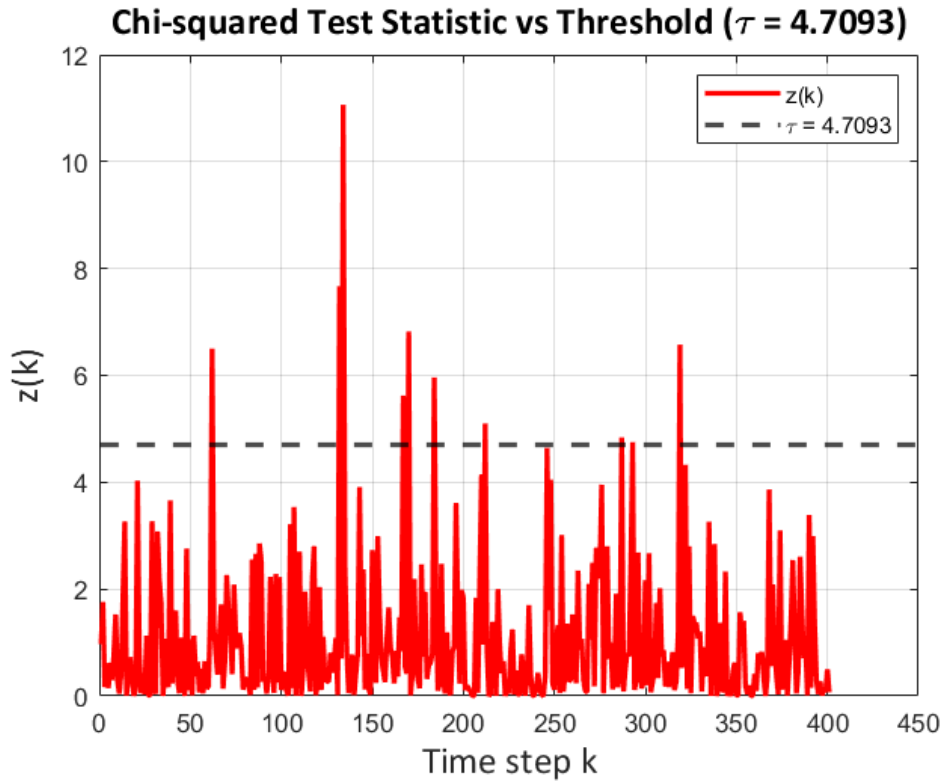


Figure 4: Chi-squared Test Statistic

Plot of the test statistic $z(k)$ vs the computed threshold $\tau = 4.70934$. Spikes above the line are false positives. False Alarm Rate = 3.74%, very close to the target of 3% which is slightly over due to randomness in process noise. 750 false positives over 20,050 checks (50 trials $\times$ 401 time steps) is within acceptable range. No excessive false alarms, hence system is not overly sensitive.

Part 2

This part evaluates the performance of the previously designed χ^2 detector under sensor attack conditions. The goal is to analyze the detection performance in the presence of a replay attack on the sensor, and to quantify how the detector responds to abnormal conditions using key detection and performance metrics.

Design Steps and Reasoning

The system was subjected to a sensor replay attack from time step 150 to 250 during each simulation. The results were averaged over 50 trials.

Parameters

Parameter	Value
Chi-squared threshold (τ)	4.7093
Significance level (α)	0.03
Degrees of freedom (v)	1
Kalman Gain (L)	[-0.0585; 0.0848]
LQI Gain (K)	[[[-0.0246, 0.0926, -0.0181]; [-0.0707, 0.4334, -0.0892]]

Evaluation Metrics (Averaged Over 50 Trials)

Metric	Value
True Positives (TP)	5050
False Negatives (FN)	0
False Positives (FP)	2692
True Negatives (TN)	12308
Detection Rate (DR)	100.00%
False Alarm Rate (FAR)	17.95%
F1 Score	78.96%
Tracking Error (Je)	4.1469

Explanation of Figures and Implications of Indices

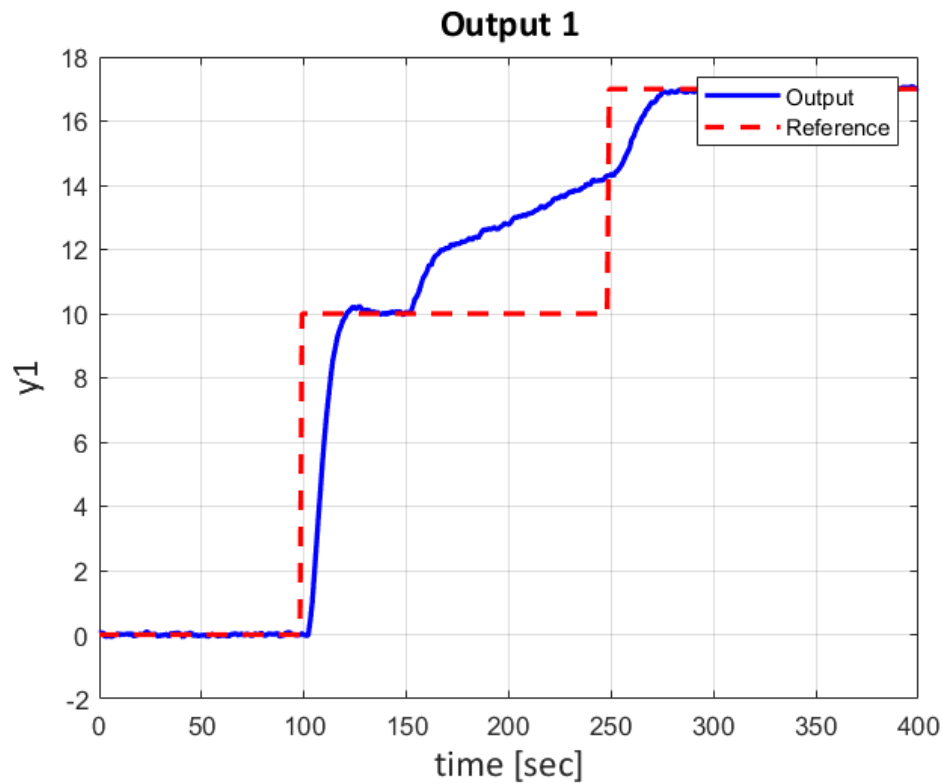


Figure 1: Output vs Reference

During the attack phase (between 150s–250s), the output significantly deviates from the reference. This deviation indicates a loss of tracking accuracy, which is directly reflected in the tracking error metric $J_e = 4.1469$. Before and after the attack, the controller tracks the reference reasonably well, showing that the attack disrupts performance locally.

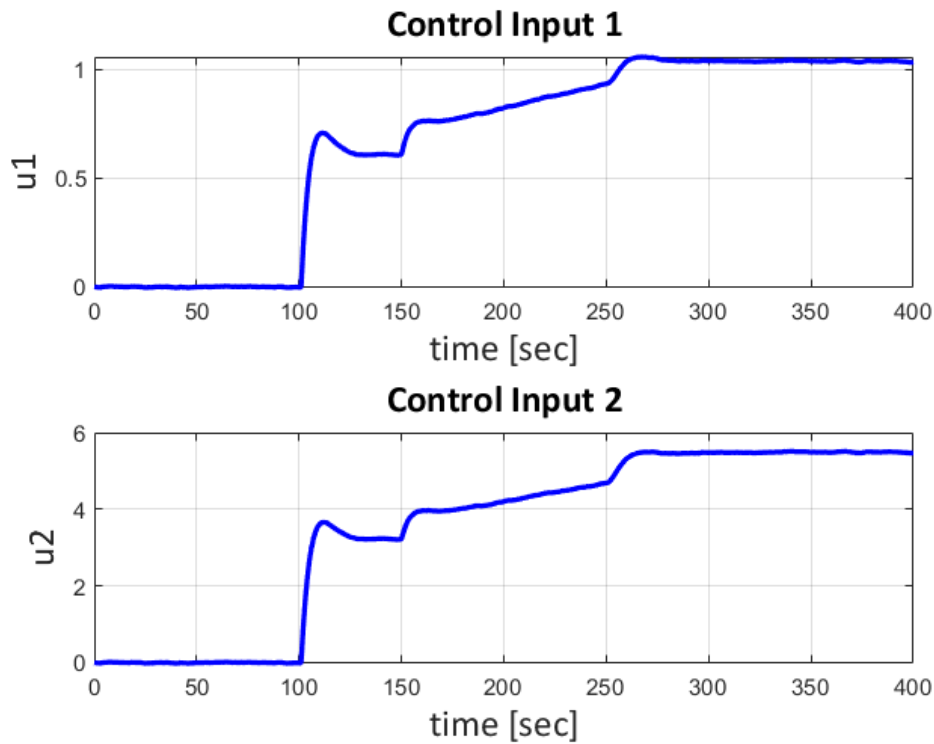


Figure 2: Control Inputs

We can see in this figure, the controller reacting aggressively during the attack period to compensate for the false sensor measurements. This shows the effort of the controller to regain stability and track the reference, even though the sensor input is compromised. It reinforces the need for a detector—without it, the controller would continue compensating blindly.

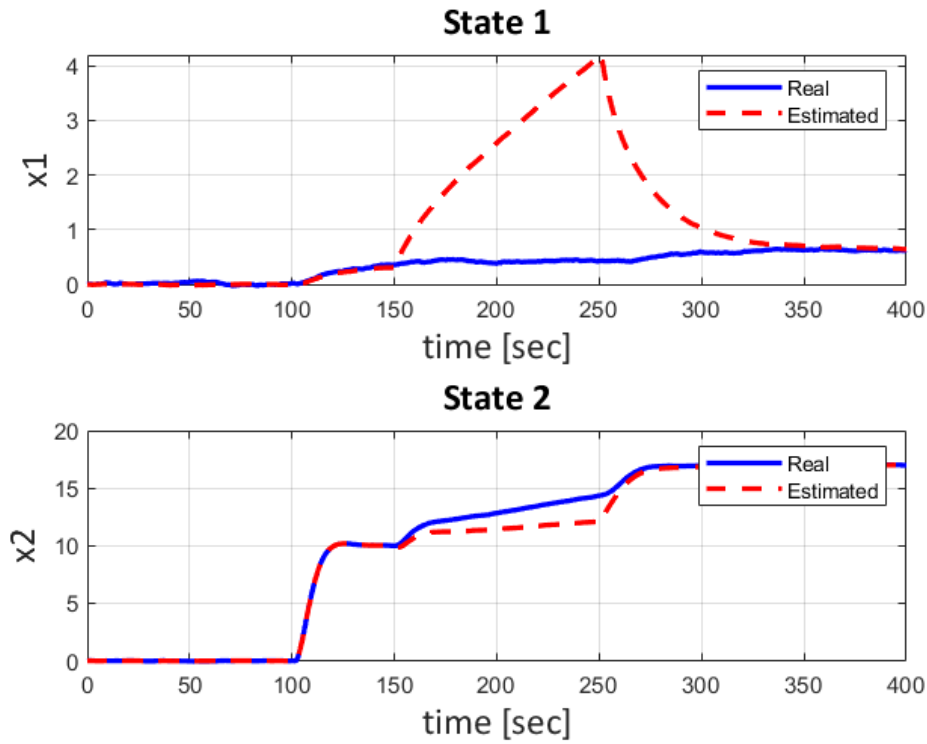


Figure 3: Real vs Estimated States

During the attack interval, the estimation diverges slightly but not drastically, showing that the Kalman Filter is still able to track fairly well despite the sensor compromise. Minor divergence impacts the detector's chi-squared residuals, which the detector uses to assess anomalies. Detection performance is still high (TP = 5050, FN = 0), meaning the estimator provides reliable inputs for detection.

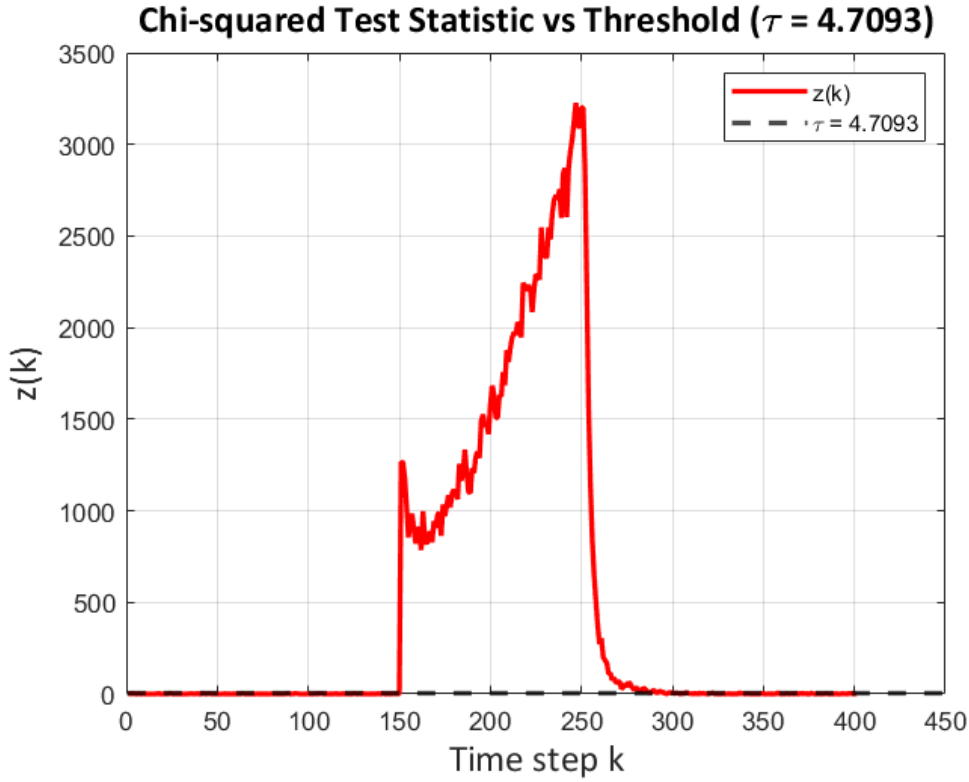


Figure 4: Chi-squared Test Statistic

The $z(k)$ values clearly exceed the threshold throughout the attack interval, indicating the detector is successfully identifying the anomaly. This results in a Detection Rate (TPR) of 100% (i.e., no false negatives). However, some $z(k)$ spikes occur outside the attack window, contributing to False Positives (FP = 2692) and a False Alarm Rate of 17.95%. Despite this, the F1 Score of 78.96% suggests a good overall balance between precision and recall.

Part 3 (Replay Attack- Equation 1.2)

This part of the assignment investigates the performance of a chi-squared anomaly detector under a coordinated replay attack (Equation 1.2) targeting both sensors and actuators. The objective is to evaluate the detector's ability to identify stealthy attacks in a networked control system.

Design Steps and Reasoning

The attack was simulated by injecting both a sensor replay and a control input assignment. The attack was executed from $t = 200$ to $t = 240$, replaying $y(t-40)$ and overriding control input u with a ramp signal on the second actuator channel.

Design Parameters

Parameter	Value
Chi-squared Test Threshold (τ)	4.7093
Sampling Time (T_s)	1 second
Attack Window	[200, 240] seconds
Replay Offset (Sensor)	40 steps
Actuator Attack (Ramp)	[0; 0.05*(t-200)]

Detection Results Over 50 Trials

Metric	Value
True Positives (TP)	83
False Negatives (FN)	1967
False Positives (FP)	187
True Negatives (TN)	9813
Detection Rate (%)	4.05
False Alarm Rate (%)	1.87
F1 Score (%)	7.16
Average Tracking Error (J_e)	2.7000

Discussion of Results and Implications

The chi-squared detector showed poor performance in detecting Replay Attack 1.2. The low detection rate (4.05%) and F1 score (7.16%) indicate that the coordinated nature of the attack made it look stealthy to the passive detector. Most attack points went undetected, evident from the high number of false negatives (FN = 1967). The chi-squared test statistic did not significantly exceed the threshold $\tau = 4.7093$ during the attack window.

Tracking performance remained relatively unaffected ($J_e = 2.7$), suggesting that while the system was under attack, the plant continued to track the reference reasonably well. This

further complicates attack detection. A more advanced active detection mechanism (e.g., watermarking) would be necessary to increase robustness against such stealthy attacks.

Relevant Plots

The following plots help visualize the effect of the attack and the performance of the detector:

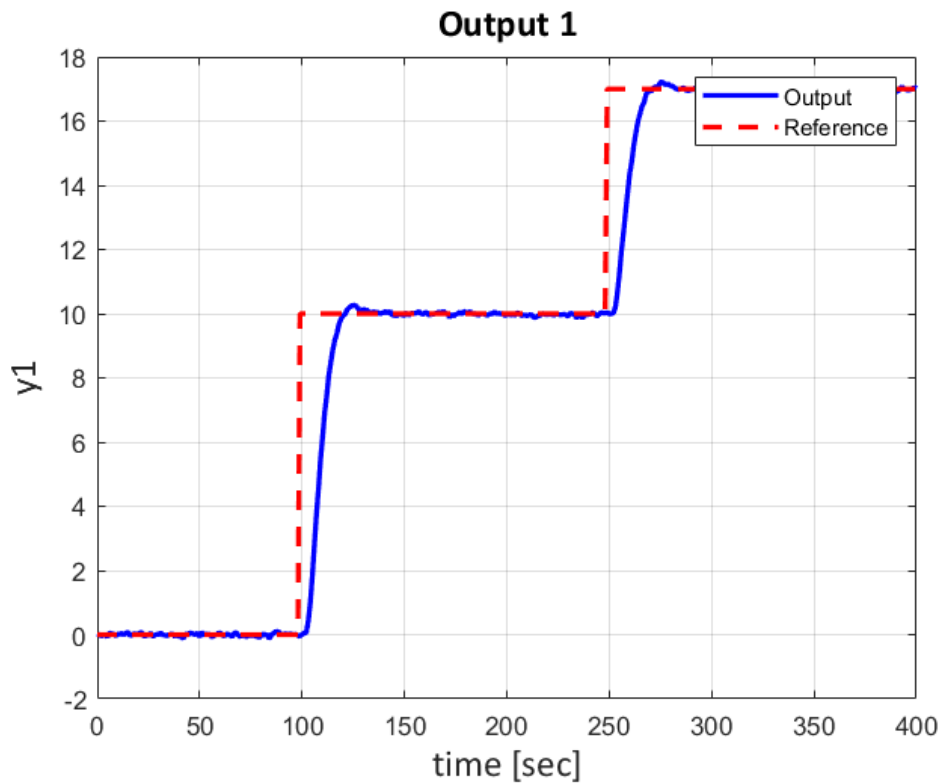


Figure 1: Output vs Reference

The plant output tracks the reference closely before and after the attack period, indicating stable control. However, the attack is not visibly reflected in the output, showcasing why detection is difficult.

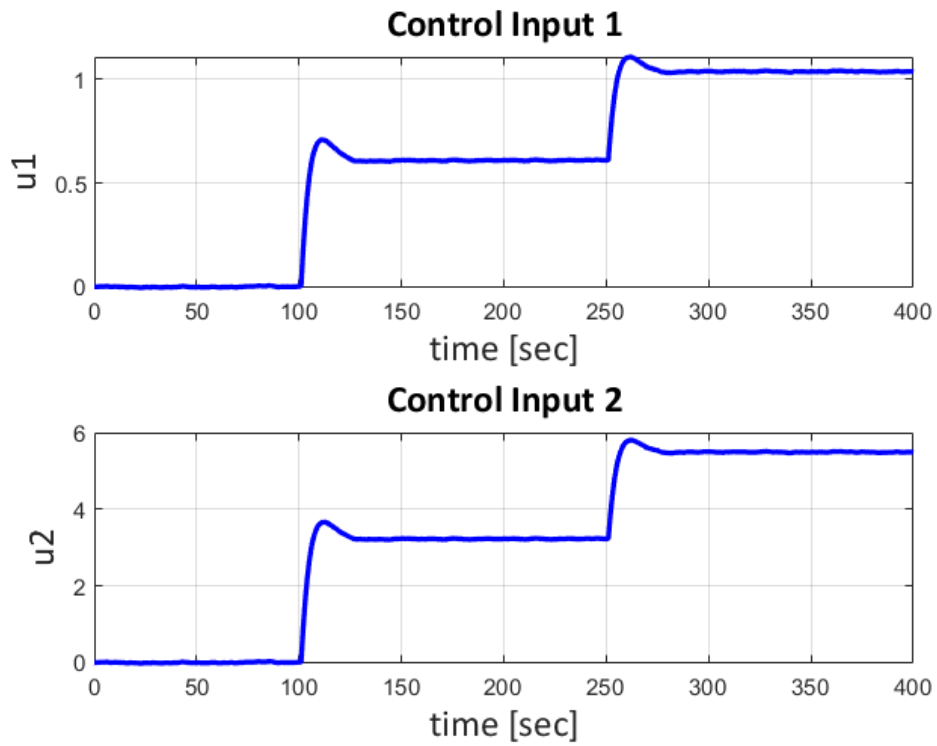


Figure 2: Control Inputs

Inputs appear normal and smooth; the ramp attack on the actuator is subtle and does not cause major anomalies. The controller does not react dramatically, implying the attack evades triggering residual changes.

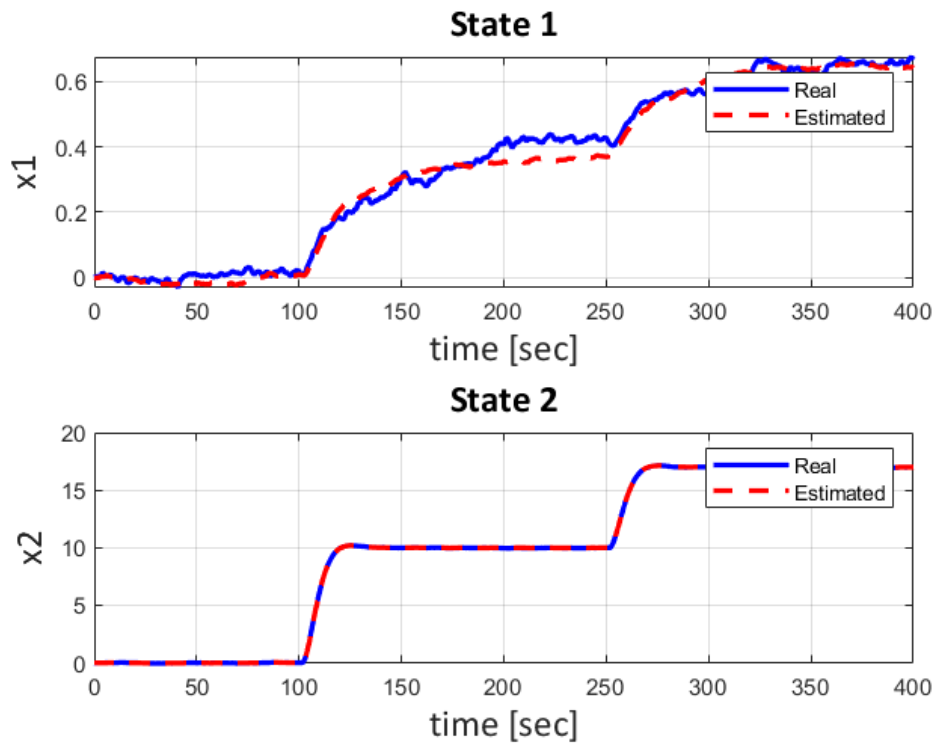


Figure 3: Real vs Estimated States

Estimated states match real states closely. No large divergence occurs during the attack, leading to undetectable residuals in the observer.

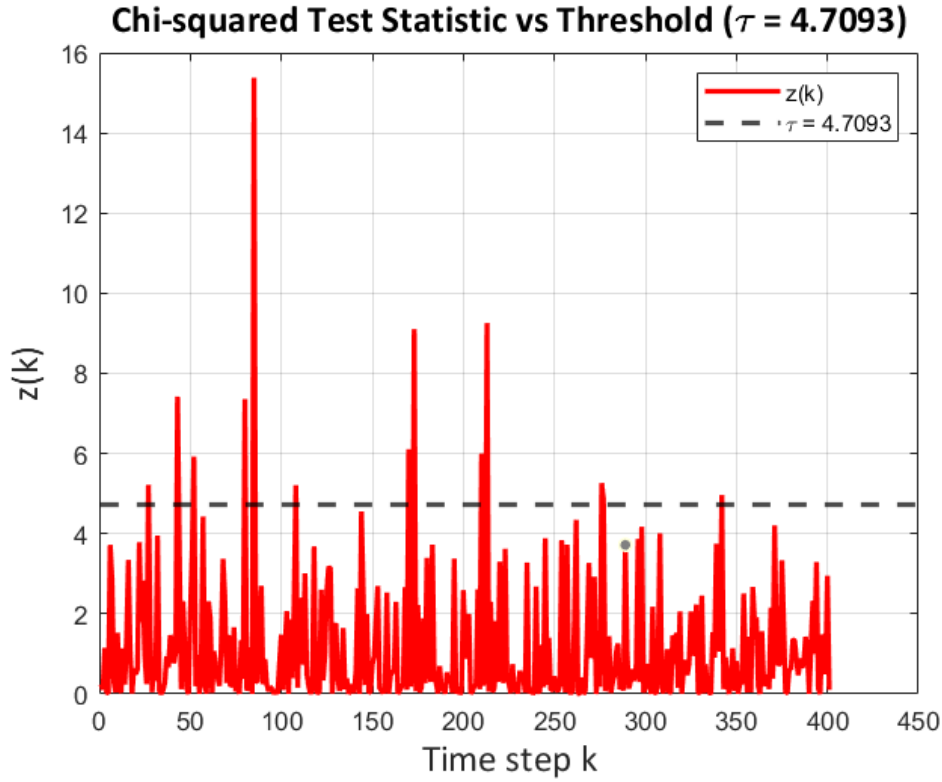


Figure 4: Chi-squared Test Statistic

The test statistic $z(k)$ remains mostly below the threshold ($\tau = 4.7093$) even during the attack window. This explains the high number of false negatives and the low True Positives (TP = 83). It indicates that the replay attack 1.2 is designed to be stealthy and successfully evades detection under a chi-squared test without watermarking. This plot validates why F1 Score is only 7.16% and Detection Rate is very low.

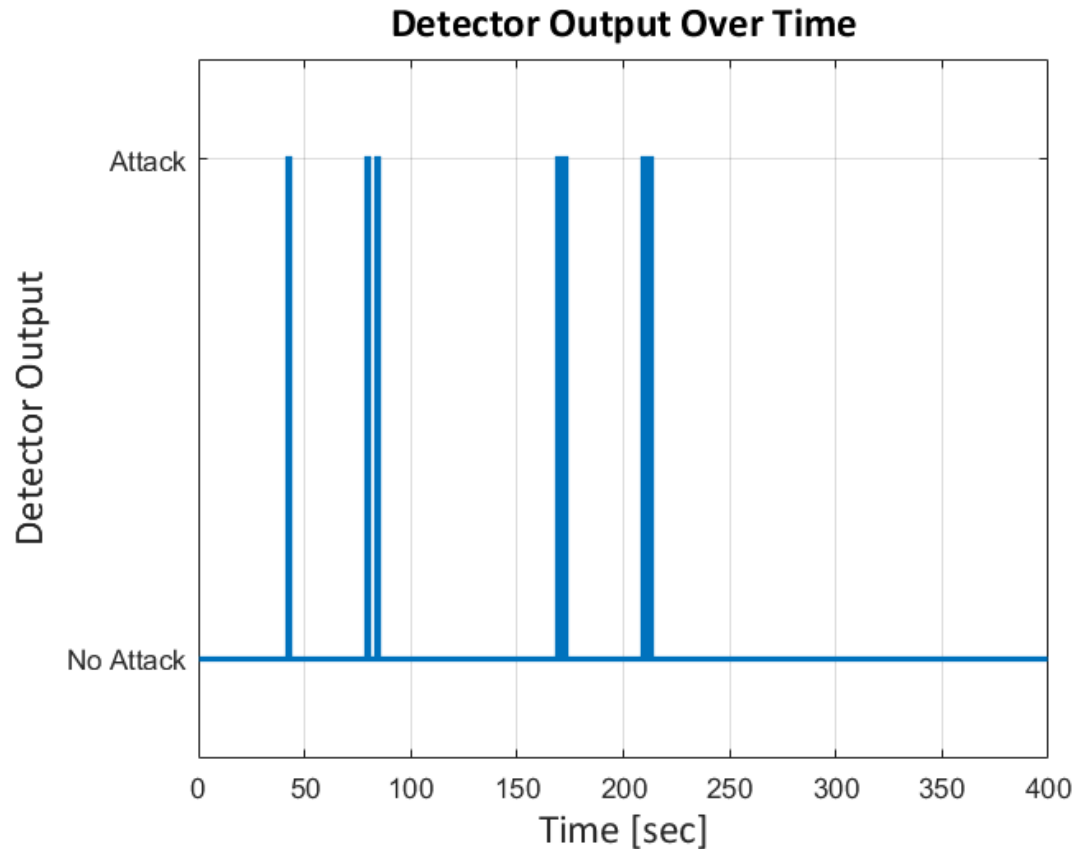


Fig5: Detector Output Over Time

Detector stays mostly in "No Attack" mode. Scattered false alarms occur early, but no consistent detection during the attack window. Confirms the weakness of passive detection against replay attacks.

Part 3 (Replay Attack- Equation 1.3)

This part of the assignment investigates the performance of a chi-squared anomaly detector under Replay Attack in equation 1.3 in a networked control system. This specific Replay Attack consists of simultaneous tampering of the control inputs and sensor measurements within the time interval [240, 280] seconds. We evaluate the detector's ability to identify these stealthy attacks by analyzing the detection rate, false alarm rate, F1 score, and tracking error.

Design Steps and Reasoning

The Replay Attack in equation 1.3 was implemented as follows:

- Sensor attack: $y'(t) = y(t) + [y(t-40) - y(t)]$ for t in $[240, 280]$
- Actuator attack: $u'(t) = [0, 0.05(t - 240)]$ for t in $[240, 280]$

Performance Results (0–280 sec)

The following indices were computed over 50 simulation runs by evaluating the detector performance up to $t = 280$ seconds:

Detection Rate (%)	False Alarm Rate (%)	F1 Score (%)	Average Tracking Error J_e
69.17	1.75	77.11	22.1691

Relevant Plots

The following plots summarize the behavior of the system under Replay Attack of Equation 1.3.

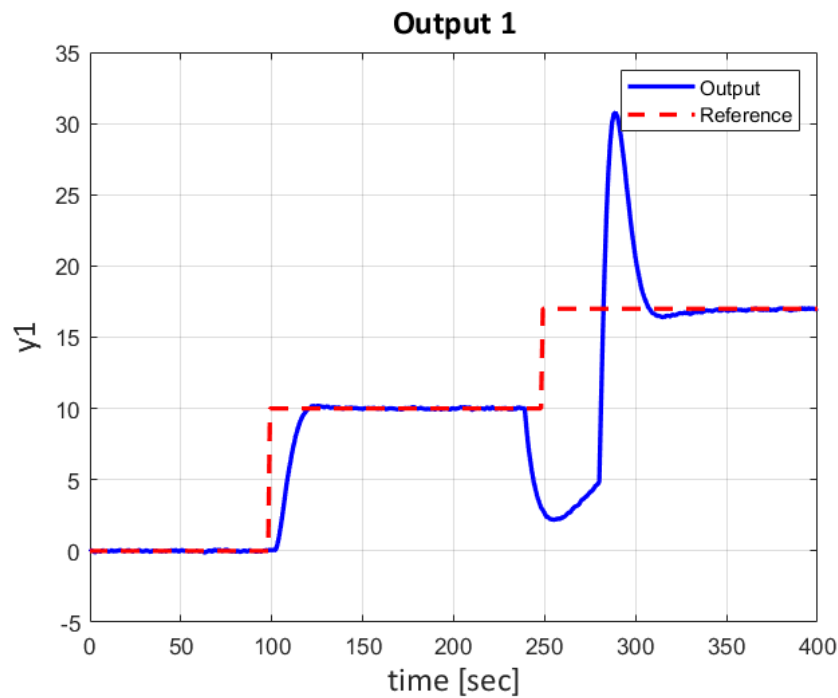


Fig1: Output vs Reference Signal

Deviation observed between 240–280 sec (attack window), where output temporarily diverges from the reference due to the attack on the sensor and actuator. The system recovers and resumes accurate tracking after the attack ends. Attack has significant short-term performance impact, but the system regains stability.

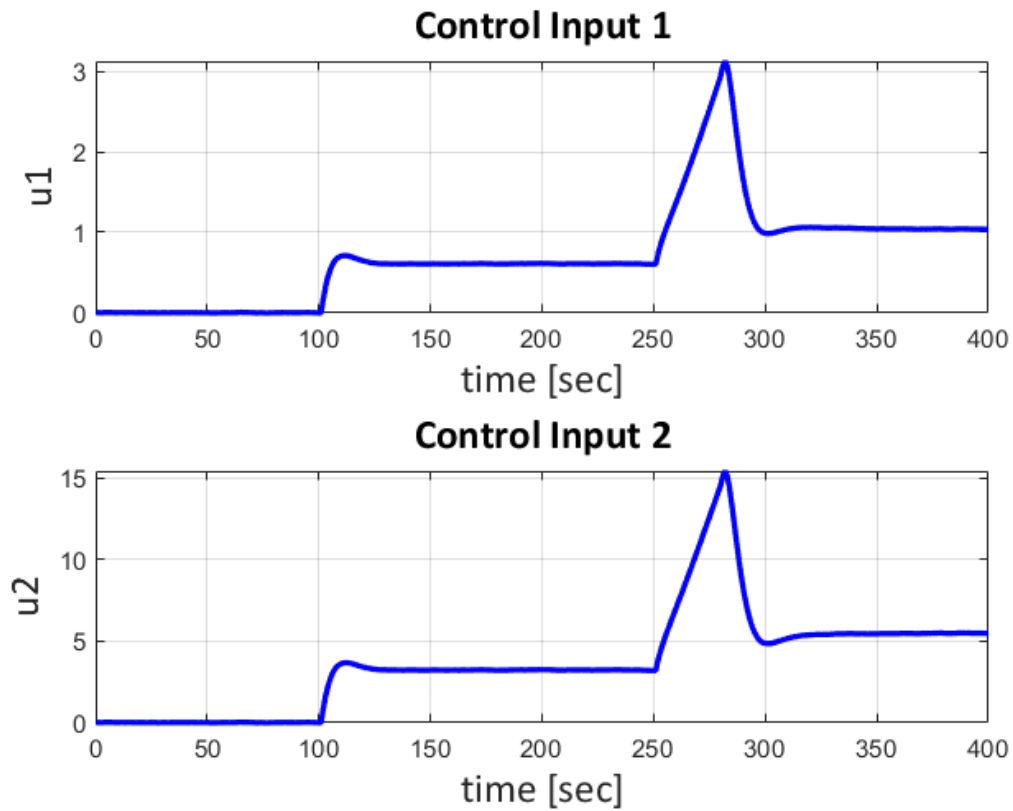


Fig2: Control Inputs

A sharp rise in both control inputs is seen during the attack window (as the system tries to compensate for misleading state estimates). This reactive behavior confirms the controller is responding to the manipulated sensor/actuator data. Control effort increases significantly during attacks, indicating control input disturbance.

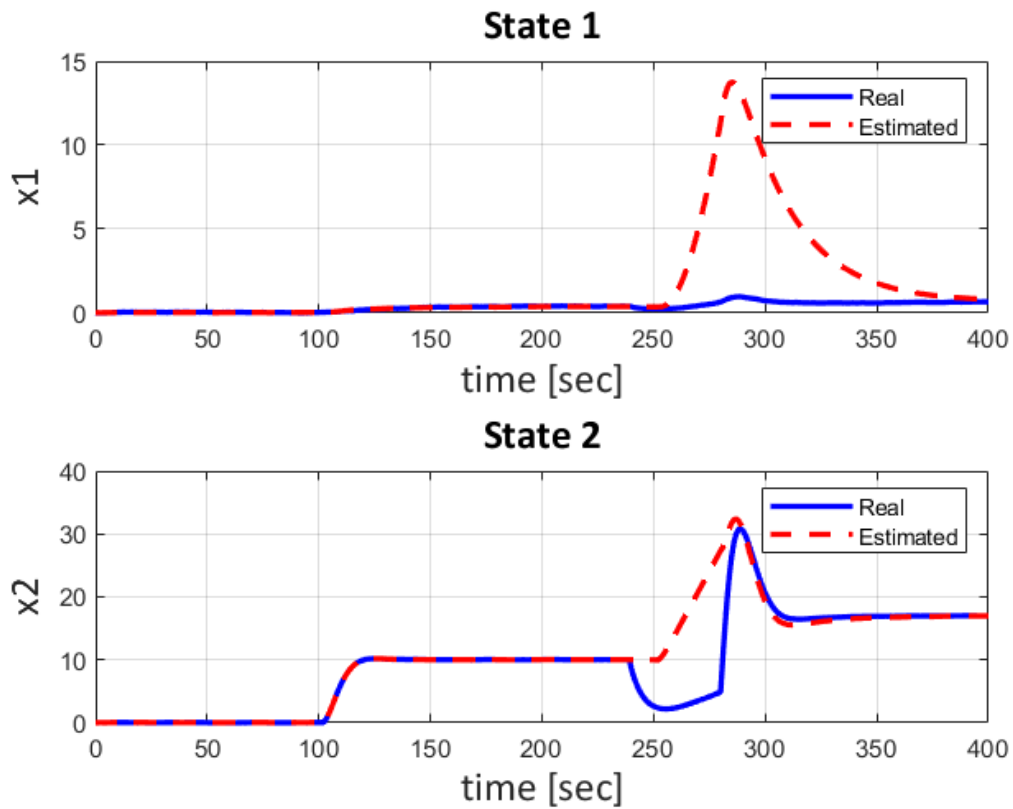


Fig3: Real vs Estimated States

During the attack, $\hat{x}$ (estimated) diverges significantly from x (real). The Kalman filter, unaware of the attack, relies on falsified sensor data, causing incorrect state estimation. Replay attack compromises the observer, leading to incorrect feedback for control.

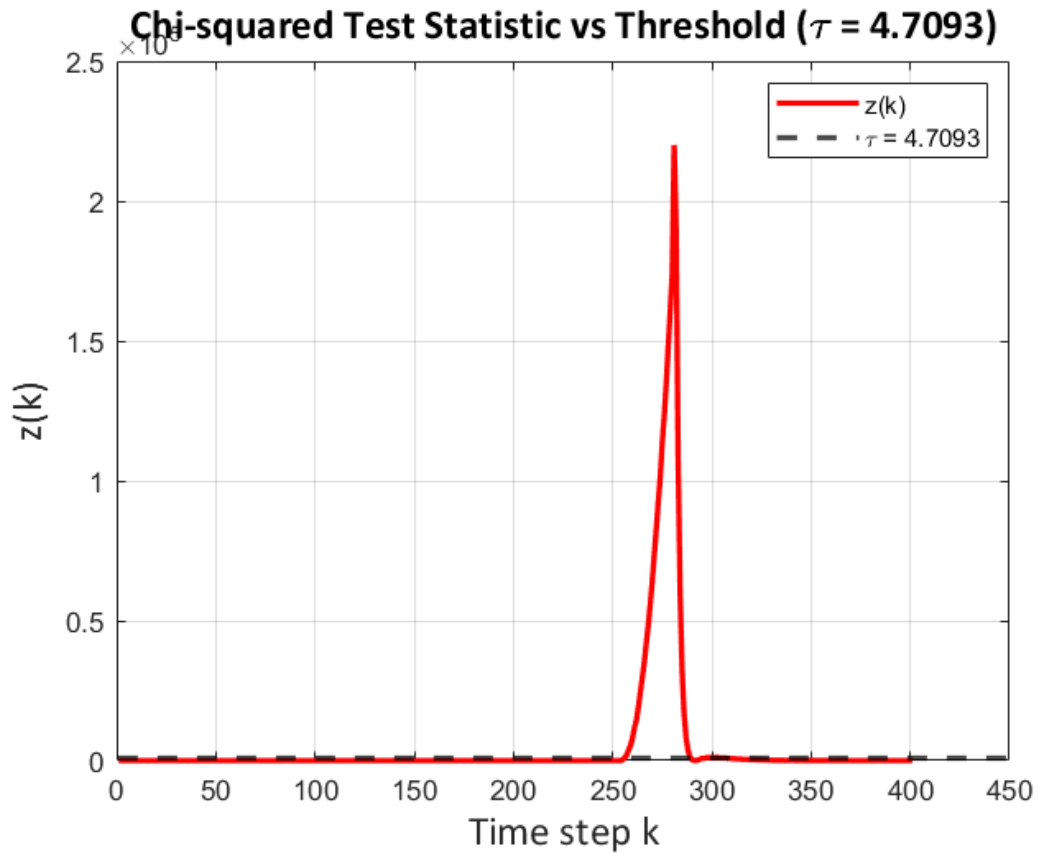


Fig4: Chi-squared Test Statistic vs Threshold

A strong peak is observed during the attack window (240–280 sec), indicating high residuals due to inconsistencies. This peak exceeds τ , triggering attack alarms. The detector successfully detects the attack during the critical window.

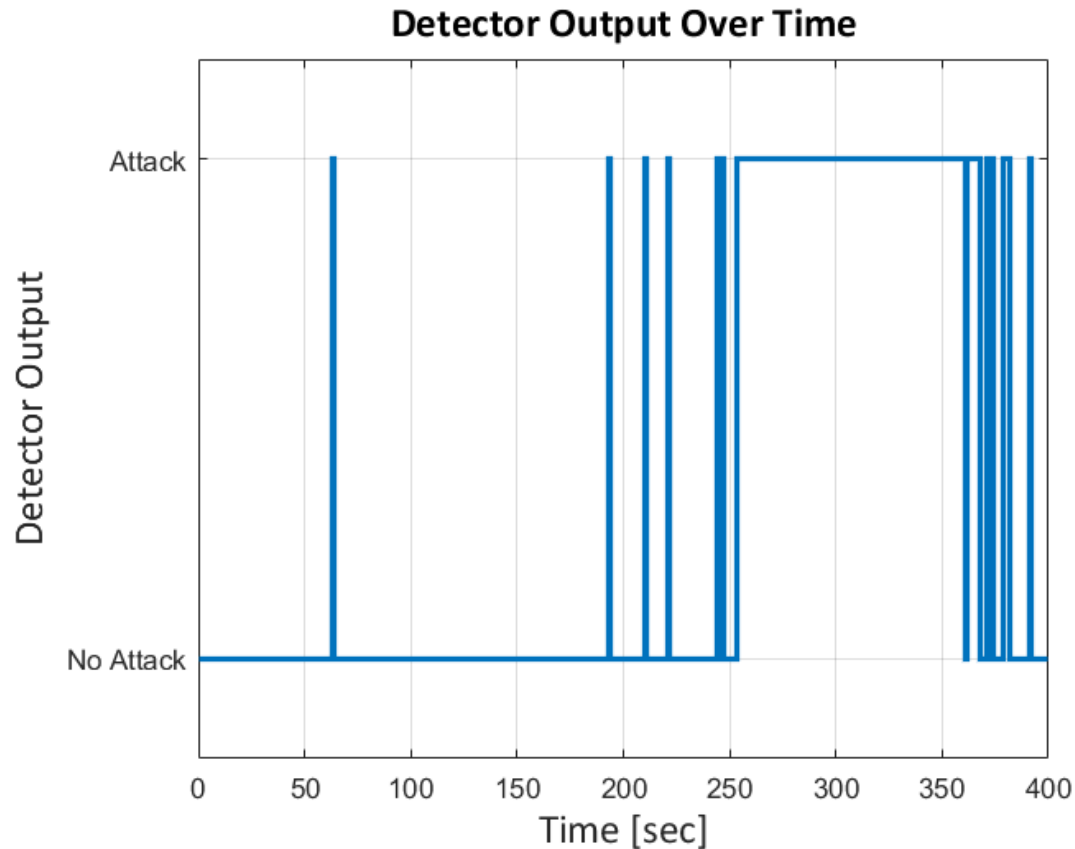


Fig5: Detector Output Over Time

1 = Attack detected, 0 = No attack. High detector activity during the attack window confirms the detection system's responsiveness. A few scattered detections before and after the attack may represent false alarms. The detector correctly identifies the attack window and shows low false alarm outside.

Implications of the Results

The detection rate of 69.17% reflects that the chi-squared detector successfully identified a majority of attack samples during the active window. The low false alarm rate (1.75%) confirms that the detector rarely triggers under normal conditions, maintaining reliability. The F1 score of 77.11% reflects a fair trade-off between sensitivity and precision in detection. The average tracking error ($J_e = 22.1691$) indicates a degradation in output tracking performance during the attack interval. The spike in chi-squared test statistic clearly aligns with the attack window, validating detection effectiveness. The "Detector Output Over Time" plot also illustrates periods where the detector correctly identified anomalies.

Comparison with Replay Attack 1.2

Compared to Replay Attack 1.2, Replay Attack 1.3 resulted in significantly better detection performance. While 1.2 had a very low detection rate (~4%) due to its stealthy

manipulation of control and sensor channels, Attack 1.3 exhibits more detectable behavior. The addition of sensor replay with stronger observable effects improves residual divergence, allowing the chi-squared detector to activate more consistently. This highlights that even small changes in the attack strategy (like replaying different segments or starting at different times) can have major implications on detectability and system safety.

Part 4

In Part 4, the goal is to implement an active anomaly detection mechanism using watermarking to detect Replay Attack in equation 1.2, defined by the specific attack input and output patterns.

Performance Without Watermarking

Metric	Value
Je(0, 200)	3.6148

Watermarking Scheme Design

The watermarking signal $\Delta u(k)$ is modeled as a zero-mean Gaussian random variable injected into the second control input:

$$\Delta u(k) \sim N(0, 0.01^2)$$

This ensures that the added noise is small enough to not violate performance constraints. Specifically, the average tracking error Je(0,200) under watermarking was validated to remain within 15% increase over the baseline value (Je = 3.6148).

Watermark Statistics:

Mean	Covariance
0	0.01^2

Performance With Watermarking (50 Trials)

True Positives (TP)	1044
False Negatives (FN)	1006
False Positives (FP)	538
True Negatives (TN)	11462
Detection Rate (drate)	50.93%

False Alarm Rate (falarm)	4.48%
F1 Score	57.49%
Average Tracking Error $J_e(0,200)$	3.6153

Relevant Plots

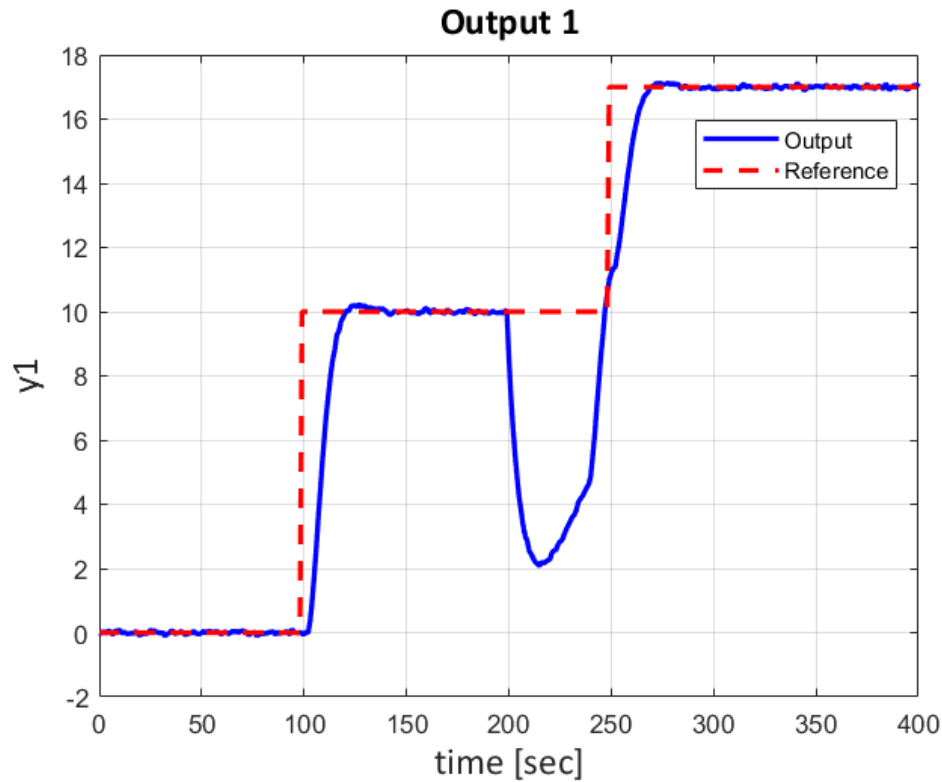


Figure 1: Output vs Reference

The output closely tracks the reference during normal operation ($t < 200$), but under Replay Attack 1.2 ($t \geq 200$), a significant deviation occurs. $J_e(0, 200) = 3.6153$ (with watermarking) confirms that the controller maintained good performance before the attack, staying within the 15% constraint over the baseline ($J_{e_noWM} = 3.6148$).

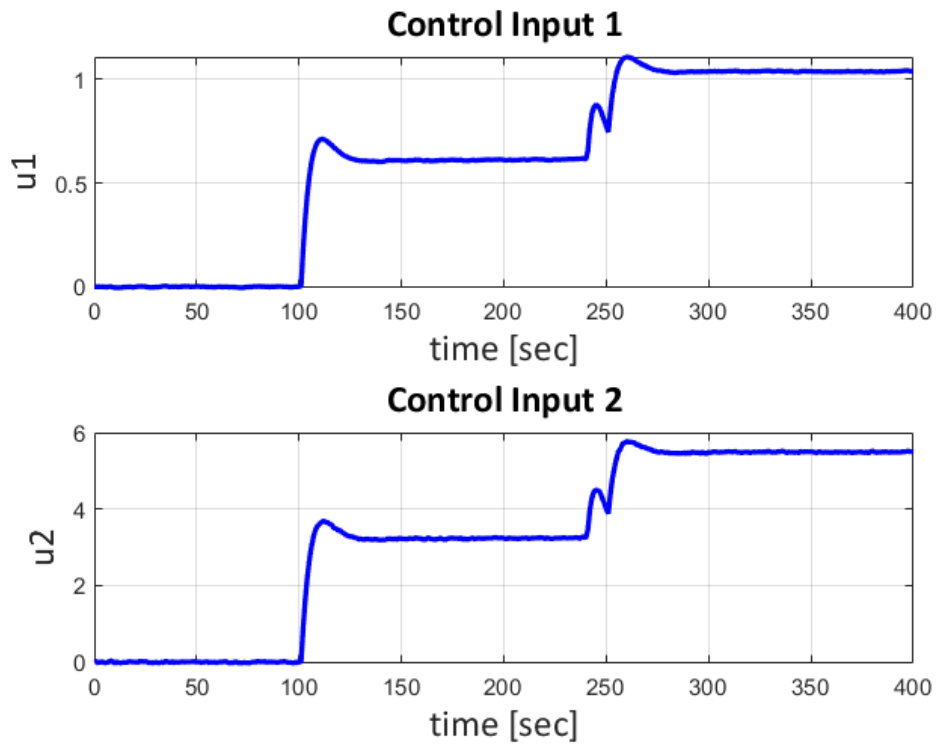


Figure 2: Control Inputs

Control signals respond appropriately to reference changes. The watermark is clearly visible as small high-frequency fluctuations in u_2 , which is where the watermark was injected. This supports that active detection is enabled, influencing the anomaly detector by making attacks statistically distinguishable.

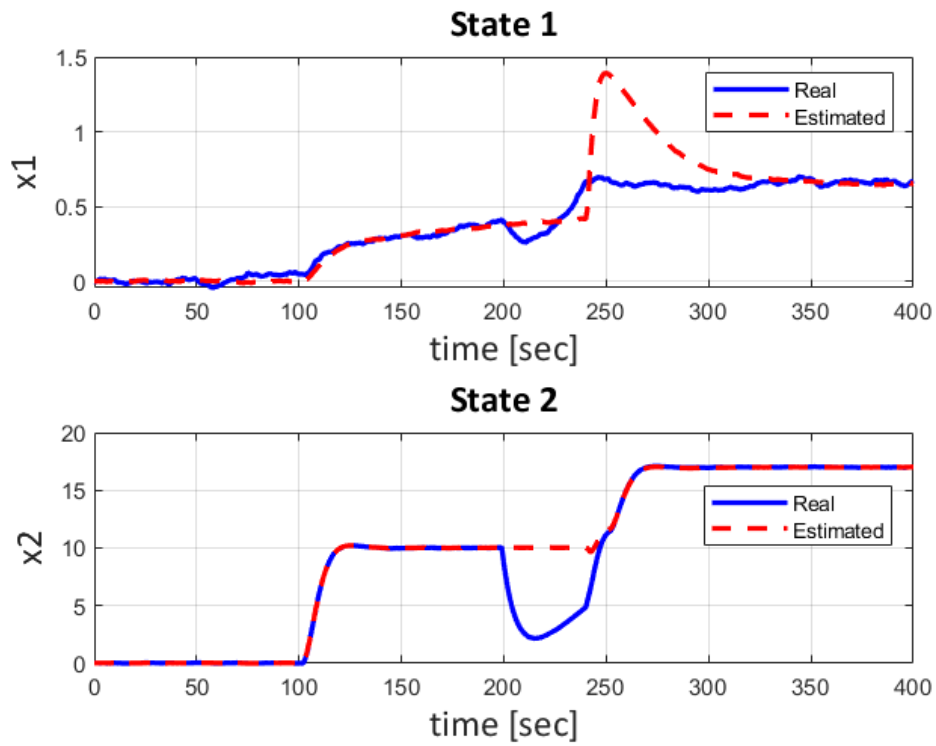


Figure 3: Real vs Estimated States

Before the attack, real and estimated states match well. During the attack, significant divergence occurs, especially in x_2 , showing that the estimator reacts to the falsified sensor data. This deviation leads to large residuals, which are used by the chi-squared detector.

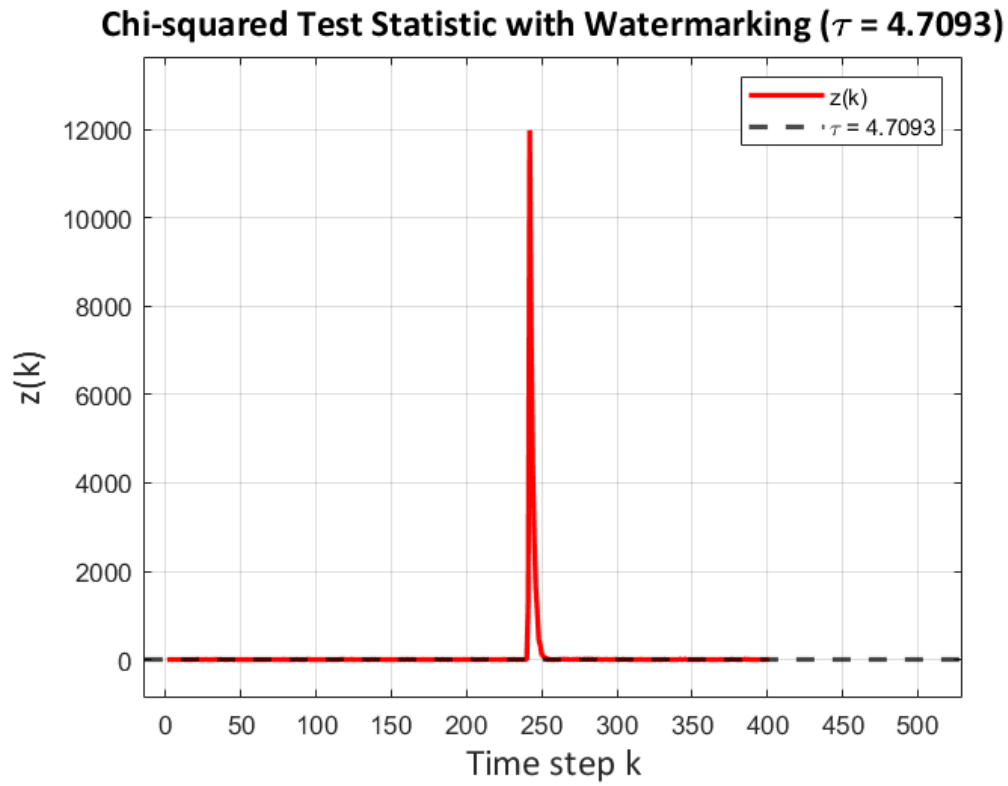


Figure 4: Chi-squared Test Statistic

A sharp spike in $z(k)$ around $t = 240$ indicates the anomaly detector reacted strongly to the attack. The spike surpasses the threshold $\tau = 4.7093$, confirming effective detection aided by watermarking.

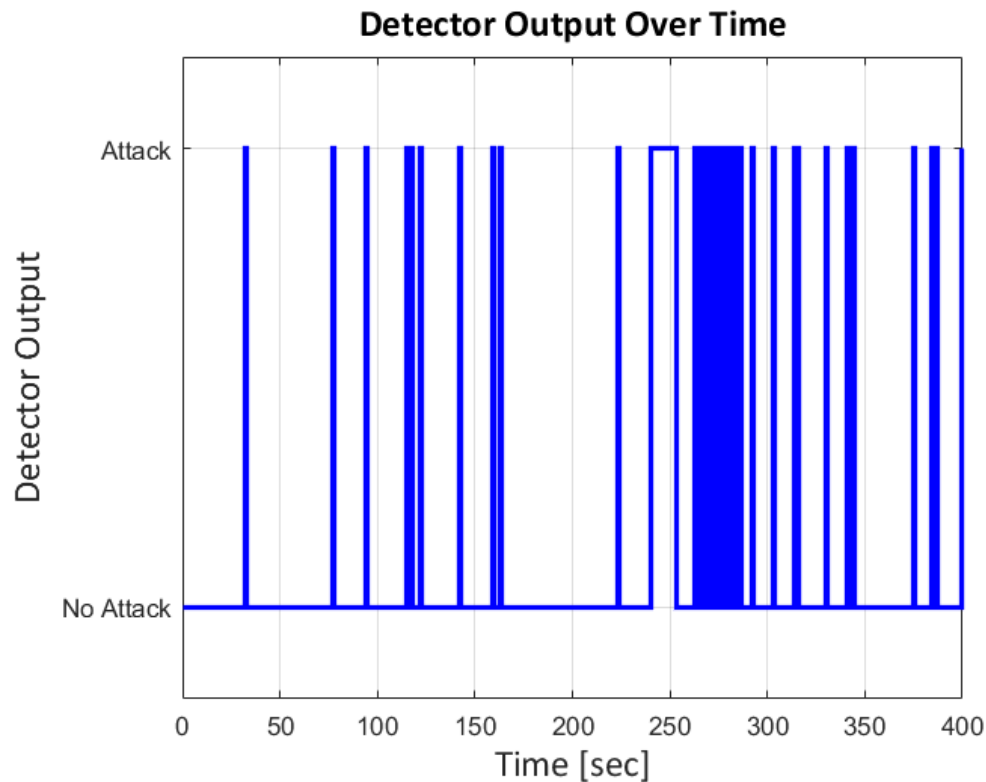


Figure 5: Detector Output Over Time

The detector identifies the attack during the replay window ($t = 200\text{--}240$) and outputs '1'. Some false positives are also observed before attack onset. Detection Rate = 50.93%, False Alarm Rate = 4.48%, and F1 Score = 57.49% reflect a moderately effective detection performance, significantly enhanced by watermarking without severely degrading tracking.

References

1. Cyber-Physical Systems Lecture Slides – INSE 6710 (Concordia University)
2. Mo, Y., & Sinopoli, B. (2009). Secure control against replay attacks. In 2009 47th Annual Allerton Conference on Communication, Control, and Computing (Allerton) (pp. 911–918). IEEE.
3. Simon, D. (2006). *Optimal State Estimation: Kalman, H Infinity, and Nonlinear Approaches*. Wiley.
4. MATLAB/Simulink Documentation: MathWorks Documentation-
<https://www.mathworks.com/help>